

Madhav Kumar

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Education

Puducherry Technological University

B.Tech in Electronics and Communication Engineering

Puducherry, India
Nov 2022 – Present

Experience

Ethical Hacking and Penetration Testing Intern – CDAC Noida

Sep 2024 – Oct 2024

Kali Linux · Network Security · Vulnerability Assessment · Metasploit

- Assessed more than five web application attack surfaces by conducting SQL injection, XSS, and DDoS simulations in controlled lab environments, documenting findings and proposing targeted mitigation strategies.
- Mapped network topologies and identified more than ten misconfigured services using Kali Linux tooling, producing a structured vulnerability report adopted by the supervising team for remediation planning.

Projects

Wercel

[Live Link](#)

Node.js · React · Docker · Cloudflare R2 · REST APIs · Microservices

- Delivered a cloud deployment platform inspired by Vercel, processing Git webhook events into fully deployed static sites in under 60 seconds via isolated Docker builds and Cloudflare R2 artifact storage.
- Exposed a REST API layer for deployment orchestration and status polling, supporting rollback to any prior build and eliminating manual intervention on failed deployments.
- Decoupled build, hosting, and dashboard into three independent microservices, enabling independent scaling and containing fault blast radius without cascading failures across the system.

CodeSM

[Live Link](#)

React · Node.js · Express.js · Redis · Docker · PostgreSQL · AWS S3

- Launched an online judge platform supporting code submissions across five or more languages with sandboxed Docker execution, enforcing CPU and memory limits on each submission to neutralize malicious payloads.
- Implemented an asynchronous job queue to decouple code execution from the API server, sustaining throughput across two or more concurrent workers without degrading response times under peak load.
- Reduced average database query load by 40% by offloading problem statements and test cases to AWS S3 and caching verdict lookups in Redis.

Documind

[Live Link](#)

Next.js · TypeScript · PostgreSQL · Vector Database · Clerk

- Developed an AI powered document intelligence platform where authenticated users query uploaded PDFs conversationally, with answers grounded in source content via a two stage retrieval augmented generation pipeline.
- Designed a chunking and embedding pipeline that indexes document segments into a vector database with metadata filtering, scoping retrieval per user and reducing irrelevant context passed to the language model.
- Enforced document isolation across sessions with Clerk authentication, ensuring every response is traceable to a cited source passage within the user's private corpus.

Chatterly

[Live Link](#)

Next.js · Socket.IO · Redis · PostgreSQL · Prisma · NextAuth

- Constructed a group messaging platform with room based event routing and automatic room joining across multiple concurrent rooms, delivering instant message history on reconnect via cached recent messages.
- Engineered an asynchronous write pipeline using Redis message broadcasting to sync events across server instances and flush buffered messages to PostgreSQL, achieving horizontal scalability without message loss.
- Secured room access control for each user with NextAuth and a session schema managed by Prisma, maintaining persistent identity across tab refreshes and network reconnections.

Skills

Languages: JavaScript, TypeScript, Node.js, C/C++, Python, Shell Scripting

Frameworks and Tools: Express.js, Nest.js, Prisma, Docker, Git, CI/CD Pipelines

Databases and Caching: PostgreSQL, MongoDB, MySQL, Redis

Cloud and DevOps: AWS (EC2, S3, Deployment), Cloudflare R2, Linux, Git Workflows

Core Concepts: Data Structures and Algorithms, System Design, Scalability, Asynchronous Programming, REST APIs

Achievements

- Secured 2nd place among more than 30 teams at JIGYASA 2025, a national level technical symposium at Puducherry Technological University.
- Solved more than 500 DSA problems across LeetCode and Code Chef, attaining a rating above 1,400 on Code Chef.